Zoom In, Scale Out: Metric Depth Models Do Not Respect a Digital Zoom

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Abstract

Metric depth models must infer the camera from pixels to output metres. We use a digital zoom, a crop followed by a resize, as a free and exact test of whether they do: the focal length is multiplied by the zoom factor s and the depth of every surviving pixel is unchanged. Two exponents summarise a model's response: its depth scales like $s^{-\alpha}$ (ideal $\alpha = 0$) and its predicted focal like s^{β} (ideal $\beta = 1$). Neither needs ground truth. Across 800 images from KITTI, NYUv2 and DIODE, the widely used Depth Anything V2 has $\alpha = 0.95$, close to reading depth from apparent size alone. Models that predict their own camera close only part of the gap ($\alpha \approx 0.31$–0.36) and under-scale the focal ($\beta = 0.45$–0.65); Depth Pro lies on the line $\alpha + \beta = 1$ that apparent-size reasoning predicts (median 0.97 per image). We explain this failure mode with a zoom–squash ambiguity: a zoomed photo is pixel-for-pixel the photo of the same scene compressed along the viewing axis, which size cues cannot detect. Controls show the effect is not caused by interpolation or detail loss. Finally, a self-consistency fine-tuning on unlabeled images reduces α to 0.05 on KITTI and 0.07 on NYUv2, once a blur-invariance term removes a "blurry means zoomed" shortcut that the plain objective learns; we report its cost on unzoomed and unseen images.

1. Introduction

Single-image metric depth has improved quickly [3, 10, 16, 23, 26], and these models now run on phones, robots and cars whose cameras they have never seen. To answer in metres, a model has to resolve a basic ambiguity: the same pixels fit a wide-angle view of a near scene and a telephoto view of a far one [9]. Some models take the intrinsics as input; others predict them; many simply regress metric depth from pixels. Whether the latter two groups really infer the camera, or fall back on the size of familiar objects, is hard to tell from benchmark accuracy, because test sets are dominated by a few cameras.

We propose a test that needs no labels and no new data. Cropping an image and resizing it back is an *exact* change of camera: the focal length is multiplied by the zoom factor s, the principal point moves in a known way, and every pixel that survives keeps its metric depth (Sec. 3). A model that understands the camera should therefore keep its depth and scale its focal by s. We ask three questions:

H1 Do current metric depth models keep the depth of the same scene points under a digital zoom, and scale their focal estimate by s?

H2 Do models that receive, predict, or ignore the camera fail in systematically different ways, and do these failures follow from a geometric explanation?

H3 Can the law be used as a training signal on unlabeled images without trading away accuracy on ordinary images?

Findings. For H1, no image-only model passes: Depth Anything V2 shrinks depth by about $1/s$ ($\alpha = 0.95$), and models with a camera head close only part of the gap while under-scaling the focal (Fig. 2). For H2, the failures line up with the *zoom–squash ambiguity* we derive: a zoomed image is identical to an image of the scene compressed along the optical axis, so cues that read apparent size cannot see a zoom; models that reason this way sit on the line $\alpha + \beta = 1$, and Depth Pro does so per image. Camera-conditioned models are equivariant with the right intrinsics and inherit exactly $+1$ in α with stale ones. For H3, a self-consistency objective in the spirit of [30] learns a shortcut, "blurry means zoomed"; adding blur invariance removes it and brings α near zero on the training domains, at a measurable cost on unzoomed and unseen images that we report in full.

Contributions.

1. A label-free **digital-zoom audit** for metric depth with exact geometry, two exponents for depth and focal, and controls for interpolation, detail loss, compression and crop position, including a separation of optical from digital zoom (Sec. 4).
2. A **taxonomy of failure modes** over three model groups and a model ladder up to Depth Anything 3 and VGGT, with the zoom–squash ambiguity and the $\alpha + \beta = 1$ line

as a falsifiable account of the dominant failure (Secs. 3 and 5).
3. An analysis of **when the label-free score predicts real error**, with scene-level uncertainty and per-dataset checks, including where it does not (Sec. 5).
4. An evaluation of **label-free correction**: self-consistency fine-tuning, the resolution shortcut it learns, EFT+B that removes it, a training-free fix for known zoom factors, and the accuracy cost on unseen domains (Secs. 6 and 7).

Code, splits and all raw results are released.

2. Related work

Metric depth and the camera. Single-image metric depth has to resolve the focal length, since the same pixels are consistent with a wide-angle view of a near scene and a telephoto view of a far one. Early work fed the focal length to the network [9] or made convolutions camera-aware [5]; FS-Depth [24] additionally augments the focal lengths of training sets by cropping, with depth labels. Metric3D [10, 27] maps every training image to a canonical camera and needs K at test time; Depth Any Camera [8] and ZeroDepth [7] likewise condition on the camera. A second family predicts the camera from pixels: Depth Pro [3] estimates the field of view, UniDepth [15, 16] predicts per-pixel rays, and MoGe-2 [23] recovers a metric point map together with intrinsics. A third family regresses metric depth without any camera input or output, *e.g.* the metric heads of Depth Anything [25, 26] and ZoeDepth [1]. Large training mixtures with many cameras [13] are meant to make camera inference robust. We provide a test of whether it is, and our three groups follow this taxonomy.

Depth cues. DepthCues [4] probes which monocular cues, among them relative and familiar size, large vision models pick up. Our exponent α measures how much a metric depth model leans on apparent size in particular, and the zoom–squash ambiguity explains why that cue cannot see a zoom.

Single-image calibration. Perspective Fields [11] and related methods predict the camera from local perspective cues, which is the cue our zoom–squash analysis says a model must use. We do not propose a calibration method; we measure whether depth models that implicitly or explicitly calibrate respond to the one camera change whose effect is known exactly.

Crops and equivariance. Zhong *et al.* [30] are the closest to us. They show that supervised depth and normal predictors are not equivariant to crop-and-resize, *i.e.* the prediction on a crop is not the crop of the prediction, and they restore it with an averaging step and a self-consistency loss, which also improves accuracy on NYUv2 when fine-tuning on unlabeled images. Their analysis treats the output as a map that should simply be cropped; it does not involve the camera. For *metric* models the same requirement has a physical reading: a crop is a zoom, the focal length must scale by s, and a model that reasons from apparent size violates it in a specific, predictable way. We measure this with exponents for depth and focal, explain it with the zoom–squash ambiguity, and test camera-predicting models that did not exist in 2023. Our fine-tuning (Sec. 6) is a self-consistency objective in the spirit of [30], adapted to metric depth with a frozen teacher; we do not claim the loss as new, and we compare it with a fine-tuning baseline that has no zoom term. Kim and Lee [12] study the deployment gap of camera-conditioned metric models when crop, resize or padding changes the intrinsics but the pipeline passes the stale ones, and propose a mismatch-aware camera module. Our stale-K experiment with Metric3D v2 is an instance of this setting; we include it to place camera-conditioned models next to the other two groups on one scale, not as a new finding. PlaneDepth [22] observes that resize-and-crop augmentation breaks the plane geometry assumed by self-supervised depth and corrects the planes for it. KPE [17] shows that crops introduce a perspective ambiguity in object-level 3D inference and adds the crop location as an input. Multi-crop human mesh recovery [14] enforces camera consistency between crops of one person. NERO [29] visualises non-equivariance of models along transformation orbits. Closest in spirit, recent work exposes the horizontal-orientation bias of depth models and makes them robust to camera roll [19]. Zoom differs from roll in one important way: a roll leaves the camera's intrinsics unchanged, whereas a zoom changes them, so the correct output is not "the same map, rotated" but a specific change of the metric prediction.

Label-free evaluation and adaptation. Self-Geometry [28] adapts 3D foundation models at test time with multi-view geometric constraints, and Re-Depth Anything [2] refines depth by self-supervised re-lighting. A self-consistency score is not an accuracy score: a model can be consistently wrong. We therefore always report the label-free exponents next to errors against ground truth, and test directly whether the former predict the latter (Sec. 5).

3. Digital zoom as an exact camera transform

The law. Let an image I of size $W \times H$ be taken by a pinhole camera with intrinsics $K = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}$. Cropping the box $b = (x_0, y_0, w, h)$ and resizing it back to $W \times H$ gives an image $\mathcal{Z}_b I$ that is *exactly* the image of the same

scene taken with

$$f'_x = s_x f_x, \quad c'_x = s_x\,(c_x - x_0 + \tfrac{1}{2}) - \tfrac{1}{2}, \qquad s_x = W/w, \tag{1}$$

and likewise for y (the half pixels come from pixel centres). Every pixel that survives the crop sees the same 3D point, so its metric depth is unchanged. For a centred crop with $s_x = s_y = s$ we call s the *zoom factor*. Eq. (1) needs no assumption about the scene, the model or the data; our implementation reproduces it to floating-point precision (Sec. 4).

What a correct model must do. Let a model map an image to a depth map $\hat{D}(I)$ and, optionally, a focal estimate $\hat{f}(I)$. Zoom equivariance asks

$$\hat{D}(\mathcal{Z}_b I)\big(\mathcal{Z}_b(p)\big) = \hat{D}(I)(p), \qquad \hat{f}(\mathcal{Z}_b I) = s\,\hat{f}(I) \tag{2}$$

for all pixels p inside b. Neither side involves ground truth.

Two exponents. Real models do not satisfy Eq. (2), but over $s \in [1, 3]$ we find their behaviour is summarised well by two power laws,

$$\frac{\hat{D}(\mathcal{Z}_b I)}{\hat{D}(I)} \approx s^{-\alpha}, \qquad \frac{\hat{f}(\mathcal{Z}_b I)}{\hat{f}(I)} \approx s^{\beta}, \tag{3}$$

fitted per image by least squares in log-log space through the origin (the depth ratio is the median over the pixels in b). An equivariant model has $\alpha = 0$ and $\beta = 1$. A model that ignores the zoom in its camera estimate has $\beta = 0$.

Three invariants. Unprojecting the prediction with the model's own focal gives a point cloud. Under a zoom, depth scales by $s^{-\alpha}$ and the lateral coordinate $X = (u - c)\,\hat{D}/\hat{f}$ by $s \cdot s^{-\alpha}/s^{\beta} = s^{1-\alpha-\beta}$. Hence

- $\alpha = 0$: depth is preserved;
- $\beta = 1$: the shape of the point cloud (the ratio X/Z) is preserved;
- $\alpha + \beta = 1$: lateral coordinates are preserved (not depth, not shape).

The third relation is what a model obtains if it reasons from apparent size. An object of metric width L that spans ℓ pixels is placed at $\hat{D} = \hat{f}L/\ell$. A zoom multiplies ℓ by s and $\hat{f}$ by s^{β}, so $\hat{D}$ scales by $s^{\beta-1}$, that is $\alpha = 1 - \beta$. A pixels-only model ($\hat{f}$ fixed, implicitly) that reasons from size has $\alpha = 1$: twice as large means twice as close.

The zoom–squash ambiguity. Why would a model reason from size at all, when the zoom is in plain sight? Because a zoom is invisible to size cues. A point (X, Y, Z) seen with focal sf projects to $u = sf\,X/Z + c = f\,X/(Z/s) + c$.

> *A zoomed photo of a scene is pixel-for-pixel the unzoomed photo of the same scene squashed along the viewing axis by s.*

The squashed scene has the same lateral sizes, so every object has a plausible width; only its extent *in depth* is wrong (a 4.5 m car becomes 2.25 m long at $s = 2$), and so are the angles between surfaces. Telling the two apart needs priors on 3D shape (aspect ratios in depth, right angles, the rate at which parallel lines converge), not priors on size. One consequence is a hard limit: for a fronto-parallel plane the squash changes nothing, so *no* estimator can be equivariant on such scenes without being told the camera. A softer reading would be that equivariance improves with the depth range of the scene, where the squash should be more visible. We tested this and did not find it (Sec. 5): the ambiguity explains the failure mode, but scene depth range is not what decides how much of it a model overcomes. The ambiguity also explains why the third invariant is the natural failure mode: believing the squashed scene means $\alpha = 1$ with $\beta = 0$, a point on the line $\alpha + \beta = 1$. Note that the line only guarantees lateral consistency; along it, depth and shape are wrong unless $\beta = 1$.

Models that take the camera as input. Models such as Metric3D [10, 27] receive K and predict depth in a canonical camera. With the correct K' from Eq. (1) they are equivariant by design. In practice the zoom often happens after capture (a crop in a gallery app, a screenshot, a detection crop) and the metadata still describes the original camera [12]. In this *stale-K* setting the network sees the same pixels and only the final de-canonical scale $f/1000$ differs by the factor s, so $\alpha_{\text{stale}} = \alpha_{\text{true}} + 1$ exactly, up to the model's output clipping. Our runs confirm this (mean difference over 800 images 0.999), so a camera-conditioned model with stale metadata behaves like a pixels-only model that reads apparent size.

4. The zoom audit

Models. We test seven models in three groups (Sec. 2): *pixels only* — Depth Anything V2 metric Small and Base [26] (indoor or outdoor head, matching the dataset) and ZoeDepth NYU+KITTI [1]; *predicts camera* — Depth Pro [3], MoGe-2 ViT-S [23] and UniDepthV2 ViT-S [16]; *takes camera* — Metric3D v2 ViT-S [10], run once with the correct zoomed K' and once with the stale K. All use the official weights and the authors' preprocessing (Hugging Face `transformers` or the official repositories).

Data. KITTI depth val selection [6, 20] (outdoor, LiDAR), the 654 official NYUv2 test images [18] (indoor, Kinect, Eigen crop), and the DIODE validation set [21] split into indoor and outdoor scans (laser). From each of the four

sets we draw a fixed random subset of 200 images (seed 0), identical for all models. Ground-truth intrinsics come with all three datasets.

Zooms and controls. We zoom by $s \in \{1.25, 1.5, 2, 3\}$ about the image centre (bicubic, antialiased, as in a phone's digital zoom). The zoom factor used in all fits is the exact one, f'_x/f_x from Eq. (1), which absorbs the rounding of the crop box. The prediction on the zoomed image is resampled onto the crop's own pixel grid by area averaging and compared, pixel by pixel, with the prediction on the original image (label-free) and with ground truth (for accuracy). To separate the camera change from the loss of detail, the *blur control* down-samples the image by $s = 2$ and up-samples it again: same detail loss as a $2\times$ zoom, same field of view. A subset of 50 images per set from KITTI and NYUv2 is also zoomed about the four quadrant centres, where the principal point moves out of the image centre.

Optical versus digital zoom. A digital zoom changes the camera *and* loses detail; a real (optical) zoom only changes the camera. We separate the two with a resolution trick. Treat a $2\times$ down-sampled copy of each image as the camera. Its *optical* $2\times$ zoom is the same field of view cut from the full-resolution image, with real detail; its *digital* $2\times$ zoom is a crop of the down-sampled image, up-sampled. Both have identical intrinsics and output size and are compared with the down-sampled camera's own prediction on the same pixels. We also repeat the $2\times$ zoom with bilinear instead of bicubic resampling, and compress the unzoomed image to JPEG quality 75.

Scope of the claim. For a model that does not receive the camera, a crop changes the visible context, and its output may legitimately change where the content is ambiguous (Sec. 3). The audit therefore measures how far each model is from the law and in which direction; it is not a claim that every image-only estimator can or should reach $\alpha = 0$ on every image.

Metrics. Label-free: the exponents α and β of Eq. (3), and the depth inconsistency at $2\times$, $\mathrm{DIE}_2 = |\operatorname{median}_p \log \hat{D}_2/\hat{D}_1|$. **With labels:** AbsRel and δ_1 on the pixels of the crop that have ground truth (capped at 80 m for KITTI, 10 m for NYUv2, 50 m and 150 m for DIODE indoor and outdoor). We report medians over images with 95% bootstrap intervals. Frames of one KITTI drive, one DIODE scene or one NYUv2 scene are not independent, so the intervals in the main tables resample whole scenes (cluster bootstrap); paired comparisons use Wilcoxon signed-rank tests with Holm correction. We also report a shape error, the median absolute deviation of $\log \hat{D}_s/\hat{D}_1$ from its median, which is zero when a zoom changes depth by a single global factor.

Sanity check of the pipeline. Projecting random 3D points with K, applying the crop-and-resize to the pixel coordinates, and projecting with K' agree to $< 10^{-3}$ px for all zooms and crop centres, and a constant depth map survives the zoom/unzoom round trip exactly (notebook 01 in the code release).

5. Results: the zoom audit

No image-only model respects the zoom (H1). Table 1 and Fig. 2 summarise 800 images per model. The pixels-only models shrink depth almost like $1/s$: $\alpha = 0.95$ [0.83, 1.12] for DA-V2-S, 0.85 for DA-V2-B and 0.61 for ZoeDepth. A $3\times$ zoom thus moves a scene to about 0.35 of its predicted distance. Models that predict their camera do better but are far from equivariant (α between 0.31 and 0.36), and their focal estimate grows by only 1.63–2.05 at $3\times$ instead of 3. Metric3D with the correct K' is close to the law ($\alpha = -0.07$, a slight over-correction), and with the stale K its exponent is $\alpha = 0.93$, larger by 1.000 per image as derived in Sec. 3. The zoom costs accuracy accordingly: on the same pixels, the AbsRel of DA-V2-S grows from 0.336 unzoomed to 0.652 at $3\times$. Intervals in Tab. 1 resample whole scenes (158 in total); the image-level intervals are much narrower and would overstate certainty.

The failure follows apparent size (H2). Figure 3 places each camera-predicting model in the (β, α) plane. Depth Pro lies on the line $\alpha + \beta = 1$ (median 0.97 per image), MoGe-2 close to it (0.95) and UniDepthV2 below it (0.87). On this line a model keeps the lateral size of what it sees but not its depth, which is exactly the interpretation the zoom–squash ambiguity makes available: the zoomed image is consistent with a compressed scene in which every object keeps its width. A softer reading of the ambiguity, that scenes with a larger depth range would be easier, is not supported: the within-dataset rank correlation between α and the inter-quartile range of log depth is 0.00 for Depth Pro and 0.02 for MoGe-2. The ambiguity explains the direction of the failure, not which images escape it.

It is the camera, not the pixels. The blur control, which loses as much detail as a $2\times$ zoom without changing the field of view, moves depth by only 1.3% for DA-V2-S and at most 4.1% for any model in Tab. 1. Table 2 separates the two effects of a digital zoom directly: an *optical* $2\times$ zoom with real detail changes the depth of DA-V2-S by a factor of –, the same zoom done digitally by –, while bilinear resampling and JPEG compression change it by – and –.

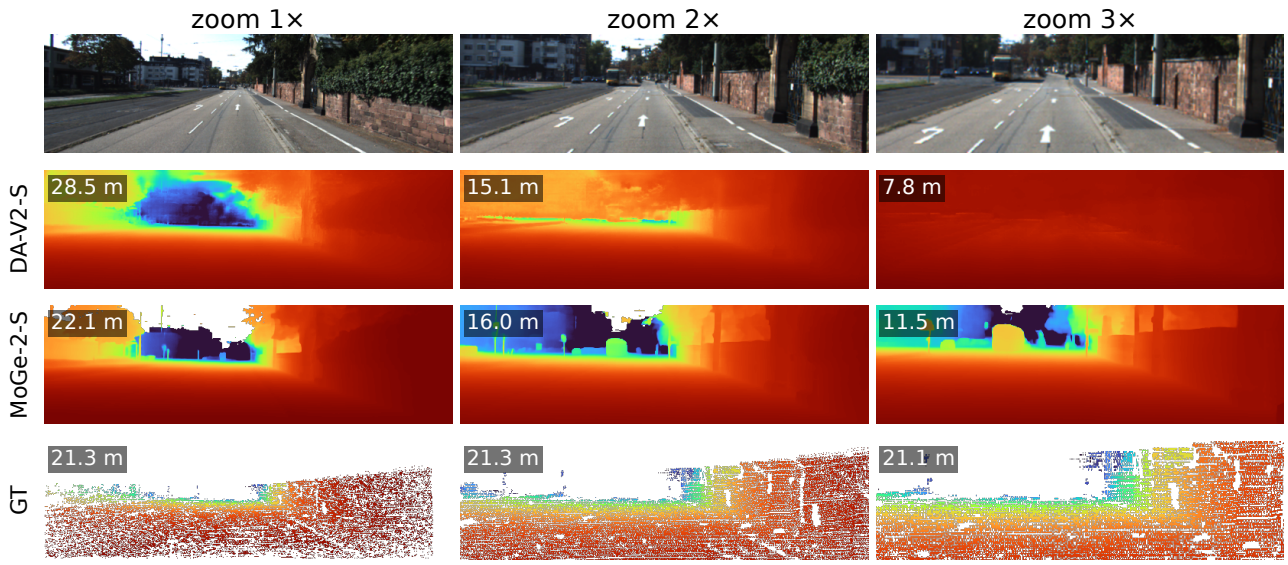


Figure 1. A KITTI image at 1×, 2× and 3× digital zoom. All depth panels share one colour scale (metres); the number is the median depth of the pixels inside the 3× crop, i.e. the same physical points in every column. Depth Anything V2 moves the scene closer as we zoom in (28.5 m to 7.8 m); MoGe-2, which predicts its camera, moves it less; the LiDAR ground truth does not move. An NYUv2 example is in the supplement.

Table 1. Zoom audit, pooled over KITTI, NYUv2, DIODE-indoor and DIODE-outdoor (800 images per model, 158 scenes). α: depth exponent (ideal 0; apparent size only: 1), median over images with a 95% interval from a scene-level bootstrap. β: focal exponent (ideal 1). $\alpha + \beta$: median per image (1 = lateral sizes kept, Sec. 3). Blur: depth change under the blur control. AbsRel at 1× on the full image, and at 3× on the crop for the unzoomed prediction ("same px") and the zoomed one.

						AbsRel		
Model	Group	α	β	$\alpha+\beta$	Blur	1×	3× same px	3× zoomed
DA-V2-S	pixels only	0.95 [0.83, 1.12]	–	–	1.3%	0.317	0.336	0.652
DA-V2-B	pixels only	0.85 [0.76, 0.94]	–	–	0.8%	0.313	0.335	0.599
ZoeDepth	pixels only	0.61 [0.53, 0.75]	–	–	0.7%	0.310	0.302	0.539
Depth Pro	predicts camera	0.31 [0.20, 0.41]	0.65	0.97	2.8%	0.310	0.293	0.459
MoGe-2-S	predicts camera	0.36 [0.30, 0.40]	0.56	0.95	3.1%	0.274	0.277	0.465
UniDepthV2-S	predicts camera	0.34 [0.23, 0.51]	0.45	0.87	4.1%	0.824	0.785	0.792
Metric3D v2-S	takes camera (true K)	-0.07 [-0.11, -0.05]	–	–	1.6%	0.142	0.148	0.358
Metric3D v2-S (stale K)	takes camera (stale K)	0.93 [0.89, 0.95]	–	–	1.6%	0.142	0.148	0.628

Bigger and newer models. Table 3 extends the audit to larger variants and to Depth Anything 3, UniK3D and VGGT (100 images per set; 50 for the DA3 and VGGT runs). On the 151 images that every model saw, size helps the pixels-only family but does not remove the effect: α falls from 0.87 (DA-V2-S) to 0.81 (B) and 0.72 (L), so the large model still returns less than half the depth at 3×. MoGe-2 is flat across sizes (0.37, 0.37, 0.35), with a focal exponent stuck near 0.55. UniDepthV2 improves with size (0.35 to 0.23, β from 0.39 to 0.68), and UniK3D-L, the newest camera-predicting model, comes closest ($\alpha = 0.17$, $\beta = 0.82$). None reaches the level of a model that is given the correct camera (Metric3D v2, $\alpha = -0.07$). Scale and newer camera heads narrow the gap; they do not close it.

Crop position and principal point. Zooming into the four image quadrants instead of the centre leaves α of the same order (supplement). For a 2× crop of an image quadrant the law moves the principal point by half the image width. Both models that predict it ignore the move: the error of MoGe-2 (0.496 of the width) and of UniDepthV2 (0.494) equals that of always answering the image centre (0.496). Off-centre crops, which phone and detection crops usually are, are outside what their camera models express.

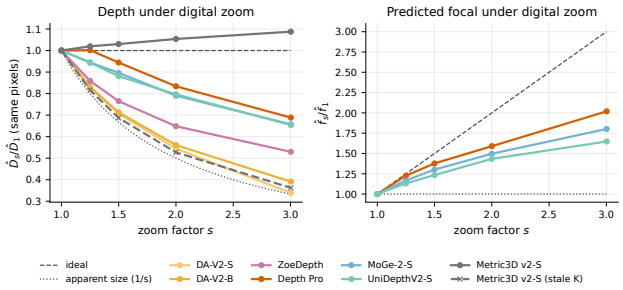


Figure 2. Depth (left) and predicted focal (right) of the same pixels as a function of the zoom factor, median over images. Dashed: the law; dotted: apparent size only.

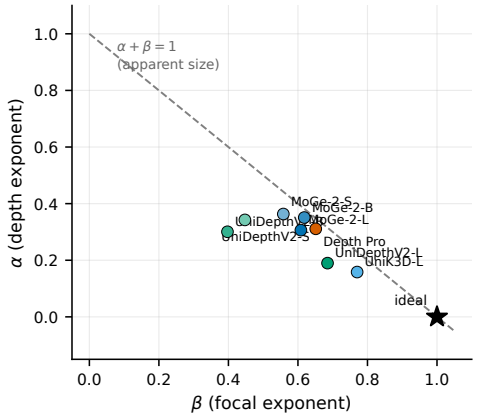


Figure 3. Camera-predicting models in the (β, α) plane. Dashed: $\alpha + \beta = 1$, where lateral sizes are preserved (apparent-size reasoning). The ideal model is at $(1, 0)$.

Table 2. Confounder controls: median depth ratio on the same pixels (1 = unchanged). Optical: real-detail 2× zoom of a half-resolution camera; digital: the same zoom by crop and upsampling; bilinear: the 2× digital zoom with bilinear resampling; JPEG: quality 75, no zoom.

Model	optical 2×	digital 2×	bilinear 2×	JPEG
DA-V2-S	–	–	–	–
Depth Pro	–	–	–	–
MoGe-2-S	–	–	–	–
Metric3D v2-S	–	–	–	–

When does the label-free score predict error? Across the 32 model–dataset pairs, $|\alpha|$ rank-correlates with AbsRel at 3× (Spearman 0.55, 95% interval [0.18, 0.86]) and within KITTI, NYUv2 and DIODE-indoor (0.98, 0.95, 0.88), but not within DIODE-outdoor (-0.38), where errors are dominated by a heavy tail of UniDepthV2. It does not predict accuracy on unzoomed images (-0.09). Part of the association is mechanical, since a shrinking depth is also a growing error, so we read it narrowly: without labels, the audit tells which model will degrade most under zoom, not which model is more accurate.

Table 3. Model ladder: larger variants and newer families. Left: each model on its own sample. Right: all models on the 151 images that every one of them saw (paired). DA3-L and VGGT normalise depth per scene, so only β is defined for them.

		own sample			shared	
Model	Group	α	β	images	α	β
DA-V2-S	pixels	0.95	–	800	0.87	–
DA-V2-B	pixels	0.85	–	800	0.81	–
ZoeDepth	pixels	0.61	–	800	0.60	–
Depth Pro	pred. K	0.31	0.65	800	0.24	0.74
MoGe-2-S	pred. K	0.36	0.56	800	0.37	0.53
UniDepthV2-S	pred. K	0.34	0.45	800	0.35	0.39
Metric3D v2-S	takes K	-0.07	–	800	-0.07	–
DA-V2-L	pixels	0.73	–	400	0.72	–
MoGe-2-B	pred. K	0.35	0.62	400	0.37	0.55
MoGe-2-L	pred. K	0.31	0.61	400	0.35	0.55
UniDepthV2-B	pred. K	0.30	0.40	400	0.31	0.44
UniDepthV2-L	pred. K	0.19	0.69	400	0.23	0.68
UniK3D-L	pred. K	0.16	0.77	400	0.17	0.82

A training-free fix when the zoom is known. Phones record a digital zoom ratio, and apps know their own crops. Since α needs no labels, a user can measure it on their own photos and multiply the depth of zoomed images by $s^{\hat{\alpha}}$. We evaluate this cross-fitted: $\hat{\alpha}$ is measured on half of a dataset's images and the correction scored on the other half. For DA-V2-S at 3× the AbsRel goes from – to –, against – for the unzoomed prediction on the same pixels. A single global factor cannot undo everything: the zoomed prediction is also distorted in shape (Sec. 4), which the remaining gap reflects.

6. Equivariance fine-tuning

The law in Eq. (1) is not only a test; it is free supervision. Following the self-consistency idea of Zhong *et al*. [30], but with a metric target and a frozen teacher, for a pixels-only model $\hat{D}_\theta$ we keep a frozen copy $\hat{D}_0$ (the teacher) and minimise, on unlabeled images I,

$$\mathcal{L} = \left\| \mathcal{U}_b \log \hat{D}_\theta(\mathcal{Z}_b I) - \log \hat{D}_0(I)\big|_b \right\|_1 + \lambda \left\| \log \hat{D}_\theta(I) - \log \hat{D}_0(I) \right\|_1, \quad (4)$$

with a zoom term (first) and an anchor (second), where b is a random box with zoom factor $s \sim \mathrm{LogUniform}[1, 3]$ and random centre, $\mathcal{U}_b$ resamples the zoomed prediction onto the crop grid, and $\cdot|_b$ crops. The teacher's prediction on the *unzoomed* image is the target for the zoomed one, so the student has to discover from the pixels that a zoom happened

Table 4. Fine-tuning DA-V2-S on unlabeled images, identical preprocessing for all variants. For each: α, AbsRel on unzoomed images, and the depth change under the blur control (the resolution shortcut). †never used in training.

	Frozen			Zhong et al.-style			Blur-only			EFT			EFT+B (ours)			EFT+B, λ=3		
Dataset	α	Abs$_{1\times}$	blur	α	Abs$_{1\times}$	blur	α	Abs$_{1\times}$	blur	α	Abs$_{1\times}$	blur	α	Abs$_{1\times}$	blur	α	Abs$_{1\times}$	blur
KITTI	1.03	0.138	1.1%	–	–	–	–	–	–	0.18	0.156	29.6%	0.05	0.175	4.9%	–	–	–
NYUv2	0.98	0.344	0.7%	–	–	–	–	–	–	0.12	0.313	4.2%	0.07	0.301	0.4%	–	–	–
DIODE-in†	0.70	0.280	1.0%	–	–	–	–	–	–	0.39	0.368	0.6%	0.37	0.400	0.4%	–	–	–
DIODE-out†	0.65	0.731	0.6%	–	–	–	–	–	–	0.39	0.877	1.5%	0.40	1.000	0.4%	–	–	–

and undo the size-based shortcut. The anchor keeps the student on the teacher for ordinary images. No depth labels and no intrinsics are used.

Setup. We tune Depth Anything V2 metric Small (24.8M parameters), the outdoor head on the frames of seven KITTI raw drives that are disjoint from the 13 drives of the KITTI val selection (3,323 images; the kernel asserts the disjointness), and the indoor head on the RGB of the 795 official NYUv2 training images. AdamW, learning rate $5 \cdot 10^{-5}$ (head) and $5 \cdot 10^{-6}$ (backbone), batch 4, 3,000 steps, $\lambda = 1$, horizontal flips, input height 392 px, mixed precision on one T4 GPU. DIODE is never used for training: it measures whether the model learned what a zoom looks like in general or only in KITTI and NYUv2. Frozen and tuned models are evaluated with identical preprocessing.

The resolution shortcut and EFT+B. A digital zoom does two things at once: it narrows the field of view and it removes detail (each output pixel is interpolated from fewer input pixels). A student trained with Eq. (4) alone can therefore learn to detect a zoom from *sharpness* instead of geometry. The blur control of Sec. 4 exposes exactly this (Sec. 7). EFT+B adds a third term that removes the shortcut:

$$\mathcal{L}_{\mathrm{B}} = \mathcal{L} + \mu \left\| \log \hat{D}_\theta(\mathcal{B}_{s'} I) - \log \hat{D}_0(I) \right\|_1, \qquad (5)$$

where $\mathcal{B}_{s'}$ down-samples by an independent $s' \sim \mathrm{LogUniform}[1, 3]$ and up-samples back. A blurred image has the field of view of the original, so its target is the unzoomed depth; detail loss alone is no longer evidence of a zoom. We use $\mu = 1$.

Baselines and ablations. Any fine-tuning changes a model, so improvements could come from the training itself. An anchor-only run is not a valid control here: the student starts as an exact copy of the teacher and sees the same input, so its loss and gradient are exactly zero and it stays the frozen model (we verified this). The *blur-only* control instead keeps the anchor and the blur term and drops the zoom term, with the same images, steps, flips and optimiser. We also report EFT+B with a three times stronger anchor ($\lambda = 3$) to probe the trade-off with accuracy on unzoomed images. All variants are scored with the same code.

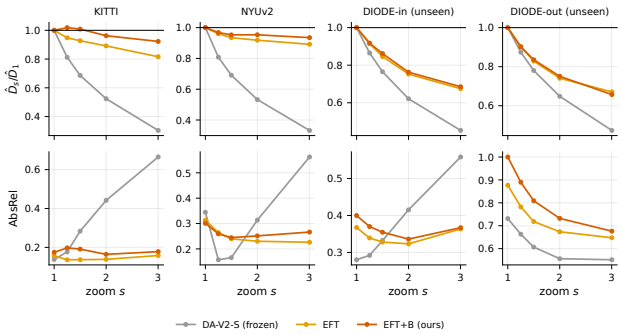


Figure 4. Depth ratio (top) and AbsRel (bottom) against the zoom factor for the frozen and fine-tuned models. AbsRel at different zooms is measured on different pixels (the crop); α, not AbsRel, is the equivariance measure: on NYUv2 the frozen model overestimates depth at $1\times$, so the shrink at moderate zoom happens to lower its error.

7. Results: correcting the models

Plain self-consistency learns a shortcut. EFT lowers α of DA-V2-S from 1.03 to 0.18 on KITTI and from 0.98 to 0.12 on NYUv2 (Tab. 4). The blur control reveals how: the tuned model moves depth by 29.6% on KITTI when the image is only blurred, against 1.1% before tuning. It has learned that a blurry image is a zoomed one.

EFT+B removes it. With the blur-invariance term the shortcut is gone (4.9% on KITTI, 0.4% on NYUv2) and α drops further, to 0.05 and 0.07. On the two datasets never used in training it is 0.37 and 0.40 (frozen: 0.70 and 0.65), so the model learns part of what a zoom looks like in general.

The cost. Equivariance is not free here. AbsRel on unzoomed KITTI images rises from 0.138 to 0.175, and on DIODE-outdoor from 0.731 to 1.000, while NYUv2 improves (0.344 to 0.301). A stronger anchor ($\lambda = 3$) and the Zhong *et al.*-style target are reported in the same table.

8. Limitations and conclusion

Limitations. A digital zoom is an exact camera change, but real zoom lenses add distortion and change focus; our optical-zoom control isolates detail loss from the camera change but is still a resampled image. For a model without camera input, a crop also removes context, so $\alpha = 0$ is a target, not a guarantee, on every image (Sec. 4). The label-free score predicts zoom-induced error, not accuracy in general, and fails on DIODE-outdoor. EFT+B trades accuracy on unzoomed and unseen images for equivariance in our setting; it is a demonstration that the law can train a model, not a drop-in replacement. Stale intrinsics in camera-conditioned models are a known deployment problem [12]; we only place them on the same scale as the other groups.

Conclusion. A crop is the simplest change of camera there is, and its effect is known exactly. Metric depth models that do not receive the camera largely ignore it: they read depth from apparent size, which a zoom cannot change, and the zoom–squash ambiguity says why. The two exponents make this visible without labels, place every model on a common scale, and point to what a correction has to avoid: a model can learn to detect a zoom from blur instead of geometry. We hope the audit becomes a routine check for any model that claims to output metres.

References

[1] Shariq Farooq Bhat, Reiner Birkl, Diana Wofk, Peter Wonka, and Matthias Müller. ZoeDepth: Zero-shot transfer by combining relative and metric depth. *arXiv:2302.12288*, 2023. 2, 3

[2] Ananta R. Bhattarai and Helge Rhodin. Re-depth anything: Test-time depth refinement via self-supervised re-lighting. *arXiv:2512.17908*, 2025. 2

[3] Aleksei Bochkovskii, Amaël Delaunoy, Hugo Germain, Marcel Santos, Yichao Zhou, Stephan R. Richter, and Vladlen Koltun. Depth pro: Sharp monocular metric depth in less than a second. In *ICLR*, 2025. 1, 2, 3

[4] Duolikun Danier, Mehmet Aygün, Changjian Li, Hakan Bilen, and Oisin Mac Aodha. DepthCues: Evaluating monocular depth perception in large vision models. In *CVPR*, 2025. 2

[5] Jose M. Facil, Benjamin Ummenhofer, Huizhong Zhou, Luis Montesano, Thomas Brox, and Javier Civera. CAM-Convs: Camera-aware multi-scale convolutions for single-view depth. In *CVPR*, 2019. 2

[6] Andreas Geiger, Philip Lenz, Christoph Stiller, and Raquel Urtasun. Vision meets robotics: The KITTI dataset. *IJRR*, 2013. 3

[7] Vitor Guizilini, Igor Vasiljevic, Dian Chen, Rares Ambrus, and Adrien Gaidon. Towards zero-shot scale-aware monocular depth estimation. In *ICCV*, 2023. 2

[8] Yuliang Guo, Sparsh Garg, S. Mahdi H. Miangoleh, Xinyu Huang, and Liu Ren. Depth any camera: Zero-shot metric depth estimation from any camera. In *CVPR*, 2025. 2

[9] Lei He, Guanghui Wang, and Zhanyi Hu. Learning depth from single images with deep neural network embedding focal length. *IEEE TIP*, 2018. 1, 2

[10] Mu Hu, Wei Yin, Chi Zhang, Zhipeng Cai, Xiaoxiao Long, Hao Chen, Kaixuan Wang, Gang Yu, Chunhua Shen, and Shaojie Shen. Metric3D v2: A versatile monocular geometric foundation model for zero-shot metric depth and surface normal estimation. *IEEE TPAMI*, 2024. 1, 2, 3

[11] Linyi Jin, Jianming Zhang, Yannick Hold-Geoffroy, Oliver Wang, Kevin Matzen, Matthew Sticha, and David F. Fouhey. Perspective fields for single image camera calibration. In *CVPR*, 2023. 2

[12] Huijun Kim and Deokwoo Lee. Robust monocular depth estimation under crop-resize-induced intrinsics mismatch. *Electronics*, 15(10):2180, 2026. 2, 3, 8

[13] Baorui Ma, Jiahui Yang, Donglin Di, Xuancheng Zhang, et al. MetricAnything: Scaling metric depth pretraining with noisy heterogeneous sources. *arXiv:2601.22054*, 2026. 2

[14] Yongwei Nie, Changzhen Liu, Chengjiang Long, Qing Zhang, Guiqing Li, and Hongmin Cai. Multiple-crop human mesh recovery with contrastive learning and camera consistency in a single image. *arXiv:2402.02074*, 2024. 2

[15] Luigi Piccinelli, Yung-Hsu Yang, Christos Sakaridis, Mattia Segu, Siyuan Li, Luc Van Gool, and Fisher Yu. UniDepth: Universal monocular metric depth estimation. In *CVPR*, 2024. 2

[16] Luigi Piccinelli, Christos Sakaridis, Yung-Hsu Yang, Mattia Segu, Siyuan Li, Wim Abbeloos, and Luc Van Gool. UniDepthV2: Universal monocular metric depth estimation made simpler. *arXiv:2502.20110*, 2025. 1, 2, 3

[17] Aditya Prakash, Arjun Gupta, and Saurabh Gupta. Mitigating perspective distortion-induced shape ambiguity in image crops. *arXiv:2312.06594*, 2023. 2

[18] Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus. Indoor segmentation and support inference from RGBD images. In *ECCV*, 2012. 3

[19] Kaihua Tang, Ziqing Xia, Xiaoxu Zheng, Xiaoxue Zhang, et al. Breaking the horizontal prior: From long-tailed orientation bias to roll-robust monocular depth estimation. *arXiv:2608.00678*, 2026. 2

[20] Jonas Uhrig, Nick Schneider, Lukas Schneider, Uwe Franke, Thomas Brox, and Andreas Geiger. Sparsity invariant CNNs. In *3DV*, 2017. 3

[21] Igor Vasiljevic, Nick Kolkin, Shanyi Zhang, Ruotian Luo, Haochen Wang, Falcon Z. Dai, Andrea F. Daniele, Mohammadreza Mostajabi, Steven Basart, Matthew R. Walter, and Gregory Shakhnarovich. DIODE: A dense indoor and outdoor DEpth dataset. *arXiv:1908.00463*, 2019. 3

[22] Ruoyu Wang, Zehao Yu, and Shenghua Gao. PlaneDepth: Self-supervised depth estimation via orthogonal planes. In *CVPR*, 2023. 2

[23] Ruicheng Wang, Sicheng Xu, Yue Dong, Yu Deng, Jianfeng Xiang, Zelong Lv, Guangzhong Sun, Xin Tong, and Jiaolong Yang. MoGe-2: Accurate monocular geometry with metric scale and sharp details. *arXiv:2507.02546*, 2025. 1, 2, 3

[24] Chengrui Wei, Meng Yang, Lei He, and Nanning Zheng. FS-Depth: Focal-and-scale depth estimation from a single image in unseen indoor scene. *arXiv:2307.14624*, 2023. 2

[25] Lihe Yang, Bingyi Kang, Zilong Huang, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything: Unleashing the power of large-scale unlabeled data. In *CVPR*, 2024. 2

[26] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything V2. In *NeurIPS*, 2024. 1, 2, 3

[27] Wei Yin, Chi Zhang, Hao Chen, Zhipeng Cai, Gang Yu, Kaixuan Wang, Xiaozhi Chen, and Chunhua Shen. Metric3D: Towards zero-shot metric 3D prediction from a single image. In *ICCV*, 2023. 2, 3

[28] Seokhyun Youn, Dahyeon Kye, Sung-Ho Bae, and Jihyong Oh. Self-geometry: GT-free and plug-and-play test-time adaptation for geometrically consistent 3D vision foundation models. *arXiv:2608.10708*, 2026. 2

[29] Zhuokai Zhao, Takumi Matsuzawa, William Irvine, Michael Maire, and Gordon Kindlmann. Evaluating machine learning models with NERO: Non-equivariance revealed on orbits. *arXiv:2305.19889*, 2023. 2

[30] Yuanyi Zhong, Anand Bhattad, Yu-Xiong Wang, and David Forsyth. Improving equivariance in state-of-the-art supervised depth and normal predictors. In *ICCV*, pages 21775–21785, 2023. 1, 2, 6